

No. of Functional Features Included in the Solution

Date	15 March 2024
Team ID	VBCUA-01
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the VBCUA project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Concept Input	Allows user to enter concept name and reference explanation.	app.py	Done	Core input feature
2	Audio Upload	Accepts WAV, MP3, and M4A audio files for analysis.	Streamlit UI	Done	Core feature
3	Speech Transcription	Converts spoken explanation into text using Whisper.	transcription.py	Done	Major AI feature
4	Semantic Similarity	Compares student explanation with reference concept text.	semantic_analysis.py	Done	Core scoring feature
5	Audio Analysis	Extracts duration, RMS energy, pause ratio, and voice energy.	audio_analysis.py	Done	Speech quality feature
6	Filler Detection	Detects filler words such as um, uh, like, basically, and actually.	filler_detection.py	Done	Fluency feature
7	Final Score	Calculates weighted score and improvement suggestions.	scoring.py	Done	Evaluation feature
8	PDF Report	Generates downloadable analysis report with scores and feedback.	report_generator.py	Done	Reporting feature

Feature Summary:

Metric	Count / Value
Total Features Planned	8

Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Concept input, audio upload, result dashboard
2	Backend / Logic	4	Transcription, similarity, audio analysis, scoring
3	Database / Storage	1	Local runtime folders for audio, waveform, and reports
4	API / Integration	3	Whisper model, Sentence-BERT, imageio-ffmpeg
5	Security / Authentication	0	Not applicable; no login or database used in current version